

Kinematics and Dynamics of Machines MEL3020

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Instantaneous Center of Velocity/ Instant Center of Velocity/ Instant Center

“At every instant during the motion of a rigid body in a plane, there exists a point that is instantaneously at rest. This point is called the instant center.”

Instantaneous Center of Velocity/ Instant Center of Velocity/ Instant Center

“Given two bodies B and C moving with planar motion relative to each other in a reference frame R, there is, in general, only one location P in the plane of motion where the co-incident points at a given instant have the same velocity with respect to the reference frame R. One coincident point is fixed to body B and the other is fixed to body C. This location is called the instant center of velocities for bodies B and C.”

Intuition of existence and location

Aronhold-Kennedy Theorem of Three Centers

“Three instant centers shared by three rigid bodies in motion with respect to one another (whether or not they are connected) all lie in the same straight line.”

Instant Center